Blind, General-Purpose Image Restoration with a 4.45-GFLOP NAFNet U-Net

A Small Generative Model for Image Denoising

Code: `https://github.com/ritchi-e/GenAI_mini`

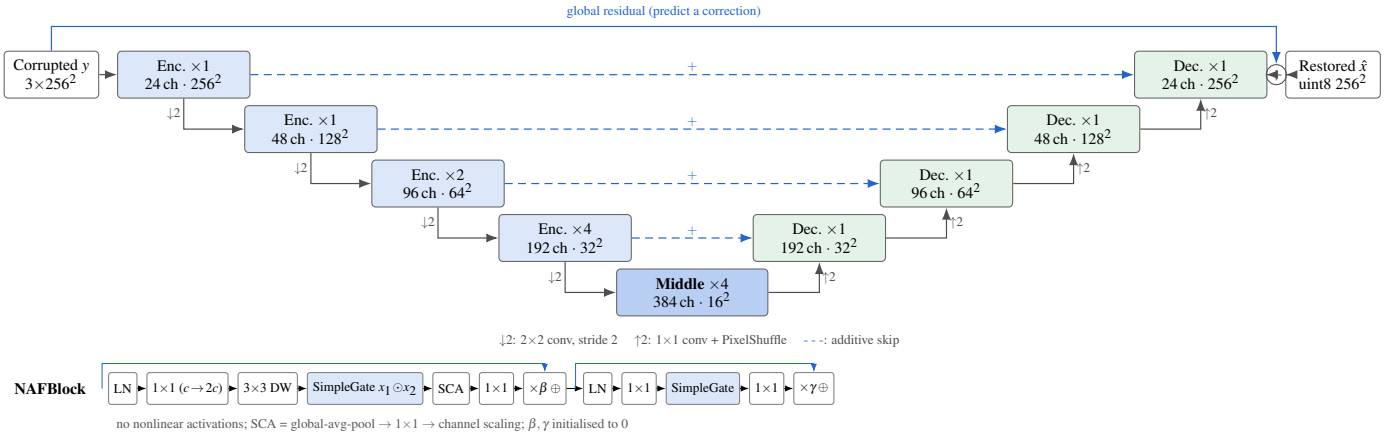


Figure 1: Submitted network (6.53 M params, **4.45 GFLOPs** fvcore / 4.52 GMACs ptflops at 1×3×256×256). Encoder/decoder blocks are NAFBlocks (bottom); most capacity sits at 1/8–1/16 resolution, where it is cheap and has the receptive field needed for blur.

1 Summary and design rationale

We submit a single-pass *conditional generative* restoration network G_θ that maps a corrupted image y to an estimate of the clean image, $\hat{x} = G_\theta(y)$, trained on pairs drawn from a broad, randomised forward corruption process $y \sim q(y \mid x)$. It is **blind**: no degradation type or level is given at test time.

Three choices follow from the scoring rule, which ranks FLOPs, PSNR, SSIM and DISTS separately and weights them equally:

- **One forward pass, no iterative sampling.** A diffusion restorer needs tens of U-Net evaluations and would lose the FLOPs axis by 1–2 orders of magnitude; our whole model costs about one step of a small diffusion U-Net.
- **Efficient blocks.** NAFBlocks lead PSNR/SSIM-per-FLOP on denoising and deblurring; at width 24 we use ∼3.6× fewer MACs than NAFNet-32 (≈16 GMACs).
- **Mean-seeking training, then a perceptual nudge.** Pixel/SSIM losses approximate $\mathbb{E}[x \mid y]$ (best PSNR/SSIM); a final DISTS stage pulls texture back toward the data manifold for the perceptual axis (Sec. 5).

2 Architecture

A 4-level U-Net (Fig. 1). Each NAFBlock is LayerNorm → 1×1 conv → 3×3 depth-wise conv → SimpleGate ($x_1 \odot x_2$) → simplified channel attention → 1×1 conv, followed by a gated FFN; both branches have learned residual scales initialised to zero, so training starts from the identity. Encoder features are *added* to the decoder (cheaper than concatenation), and a **global residual** makes the network predict only a correction, so clean regions pass through unchanged. Only conv, element-wise and mean ops are used, so fvcore/ptflops count every operation. Inference clamps the output to $[0, 1]$, multiplies by 255, rounds and returns `uint8` 256×256×3.

Table 1: Model family (same block design, measured at 1×3×256²). **small** is submitted.

Config	width	params	GFLOPs (fvcore)	GMACs (ptflops)
tiny	16	2.93 M	2.03	2.07
small	**24**	**6.53 M**	**4.45**	**4.52**
base (deeper)	32	20.9 M	11.07	11.17
NAFNet-32 (ref.)	32	29.1 M	—	≈16

3 Training data

Public research/CC datasets, sampled by domain weight:

- **Natural (0.6):** DIV2K train HR 0001–0740, cut into 384 px tiles at scales $1, \frac{1}{2}, \frac{1}{4}$; down-scaled tiles mimic whole scenes resized to 256^2.
- **Medical (0.3):** MedMNIST+ 224 px (chest X-ray, abdominal CT, breast ultrasound, retinal fundus; CC BY 4.0) and Brain-Tumor-MRI T1/T2 slices (MIT), resized to 256.
- **Low-light (0.1):** LOL normal-light images, plus synthetic darkening of any image.

45,340 source tiles; corrupted inputs are generated on the fly, so no pair is ever seen twice.

4 Degradations used during training

For every patch a random chain is applied in image-formation order, so types, severities and *combinations* vary freely:

1. **Blur** (p=.45): anisotropic **Gaussian** (σ .3–3.5), linear/curved **motion** (3–25 px), **defocus** disk (r 1–6).
2. **Chromatic aberration** (p=.2): lateral R/B scale and shift, optional per-channel blur.
3. **Resolution loss** (p=.12): down/up-resize ×1.3–4.
4. **Noise** (p=.8, a second type mixed in with p=.2): **Gaussian** (σ 1–55, colour/luma), **Poisson** (peak 5–500), **speckle**, **salt-and-pepper** (.2–10%), Rician (MRI), uniform, correlated grain, row/column banding.
5. **JPEG** q 5–95 (p=.35), bit-depth quantisation, late impulses; finally rounding to the `uint8` grid.

Every family in the brief is covered (**bold**). The extra families and random mixing target the undisclosed test corruptions. About 3% of samples are left clean to teach the model not to over-smooth. Content augmentation: flips/rotations, grayscale (p=.08) and **low-light simulation** (p=.15; gamma 1–2.2, gain .08–.6 applied to the target before corruption).

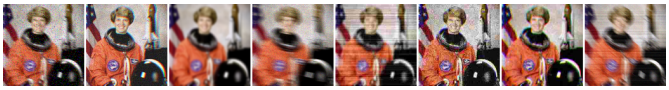


Figure 2: Eight random draws of the training degradation chain on one image.

5 Objective and schedule

$$\mathcal{L} = \mathcal{L}_{\text{Charb}} + 0.2\,(1-\text{SSIM}) + 0.05\,\|\mathcal{F}\hat{x} - \mathcal{F}x\|_1 + \lambda_{\text{D}}\,\text{DISTS}$$

Each term targets one axis: Charbonnier→PSNR, SSIM→SSIM, the FFT-L1 term→sharp high frequencies after deblurring, DISTS→perception. AdamW (β=.9, .9), lr 10^{-3}, 2k warm-up, cosine decay to 10^{-6}, gradient clipping 0.5, bf16, **EMA** 0.999 (the EMA weights are submitted). 200k iterations on **one NVIDIA A30, ≈8.2 GPU-hours**, in three stages: 0–150k at 128 px, batch 32 (4.9 h);

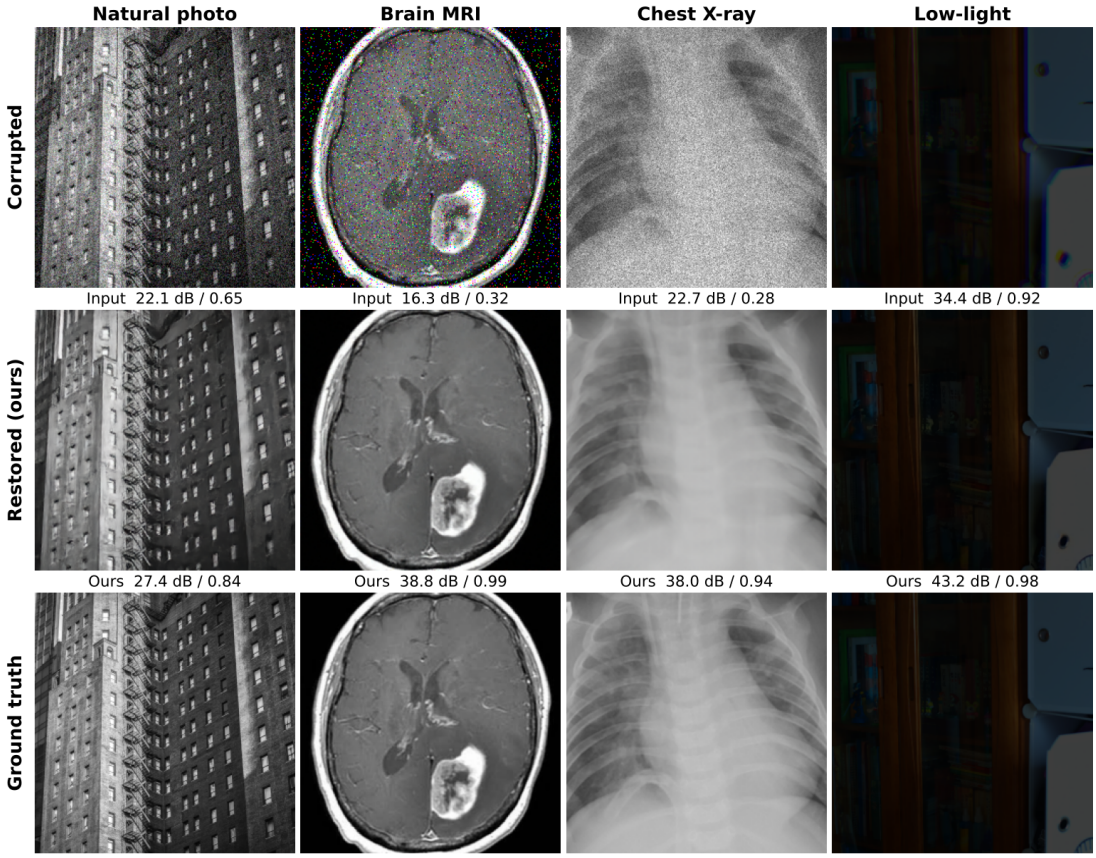


Figure 3: Validation examples from each content domain (PSNR / SSIM against the clean ground truth). Gaussian/Poisson noise, impulses and colour fringes are removed while edges and anatomical structure in the MRI and X-ray are preserved; the low-light example shows the model leaves dark scenes dark instead of brightening them.

150–185k at 256 px, batch 16, the true test size (2.0 h); 185–200k at 256 px with λ_D=0.05 (1.3 h). The model is validated every 5k iterations, and the checkpoint maximising PSNR/40 + SSIM − DISTS is kept.

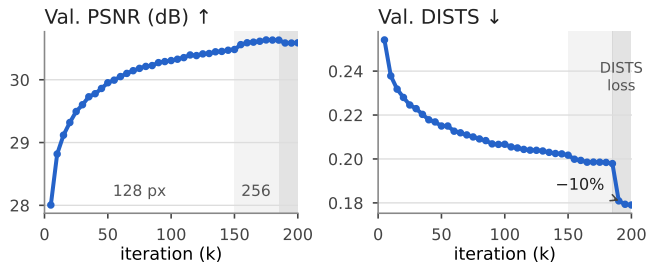


Figure 4: Validation curves (EMA weights, 119 images). The 256 px stage (150k→185k) adds +0.15 dB; the DISTS stage cuts DISTS by 10% (0.198→0.179) for −0.04 dB PSNR and −0.001 SSIM.

6 Results

Validation protocol. 119 pairs with no image overlap with training: 60 DIV2K (0741–0800), 32 MedMNIST *test*, 12 MRI *test* and 15 LOL-eval images (darkened), each with its own seeded degradation chain. Metrics match the competition: `skimage` PSNR/SSIM and official DISTS on `uint8` 256^2 images.

Table 2: Validation results of the submitted weights.

Domain	n	$\text{PSNR}_{\text{input}}$	PSNR↑	SSIM↑	DISTS↓
Natural	60	19.69	25.96	0.764	0.205
Medical (X-ray/CT/US/MRI)	44	22.10	34.08	0.891	0.136
Low-light	15	27.26	38.86	0.915	0.203
All	119	21.53	**30.59**	**0.830**	**0.179**

Out-of-pipeline check. Because the validation corruptions come from our own generator, we also wrote each brief corruption *independently* (plain NumPy/OpenCV, fixed severities) and applied it to four validation images (Table 3). The model improves every case, including multi-step mixtures, and leaves clean images untouched (53.5 dB, SSIM 0.999), so it does not hallucinate structure.

Table 3: Independently implemented corruptions (mean of 4 images; input → output).

Corruption	PSNR (dB)	SSIM
Gaussian σ=25 / σ=50	21.0→**31.7** / 15.5→**28.6**	.42→.91 / .23→.83
Poisson (peak 20)	19.8→**31.3**	.46→.90
Salt & pepper 5%	17.6→**40.7**	.46→.99
Speckle σ=0.2	25.3→**34.3**	.74→.95
Chromatic aberration (3 px)	22.8→**35.3**	.70→.95
Gaussian blur σ=2	26.7→**30.9**	.74→.90
Motion blur 15 px	25.2→**27.7**	.68→.80
JPEG q=10	26.9→**27.9**	.78→.83
Motion + Gaussian + JPEG	25.0→**27.5**	.61→.80
Clean input (identity)	—→53.5	1.00→1.00

7 Limitations

- Validation shares the training corruption generator, so test scores will likely be below 30.6 dB; Table 3 is the more conservative estimate.
- Severe motion blur and JPEG $q \leq 10$ gain least (+2.5 / +1.0 dB); at high noise, fine texture (fur, foliage) is smoothed. Haze/contrast loss is not corrected.
- Only one size was trained; `tiny` (2.0 GFLOPs) and `base` (11.1 GFLOPs) configs are provided to trade FLOPs rank against quality.

8 Submission

`weights/model.pth` ({`"state_dict"`,`"arch"`}, fp32) and `weights/model.onnx` (opset 17, max. deviation from PyTorch 4×10^{-7}). Run `python inference.py --input <img|dir> --output <dir>`: `uint8` or float in, always `uint8` $[0,255]$ $256\times256\times3$ out. FLOPs: `python count_flops.py --weights weights/model.pth`.